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1- OVERVIEW

4kw RF AMP is designed to be used for 1.0T medical MRI (Magnetic Resonance Imaging) equipment. All electronic circuits inside the RF AMP are solid state, providing long life. Therefore, unlike the other RF AMP of electronic tube type, there is no need for exchanging electronic tubes regularly, nor for complex fine-tuning.

This product has the following characteristics.

- A) Working frequency of 42.68MHz 225KHz
- B) Pulse output only--Can not be used under continuous wave
- C) Peak output of 4kw, Unblank pulse width of 30m sec. max., Duty of 60% max.
- D) RF output noise level under blank condition of less than 160dBm/Hz

2- CONFIGURATION

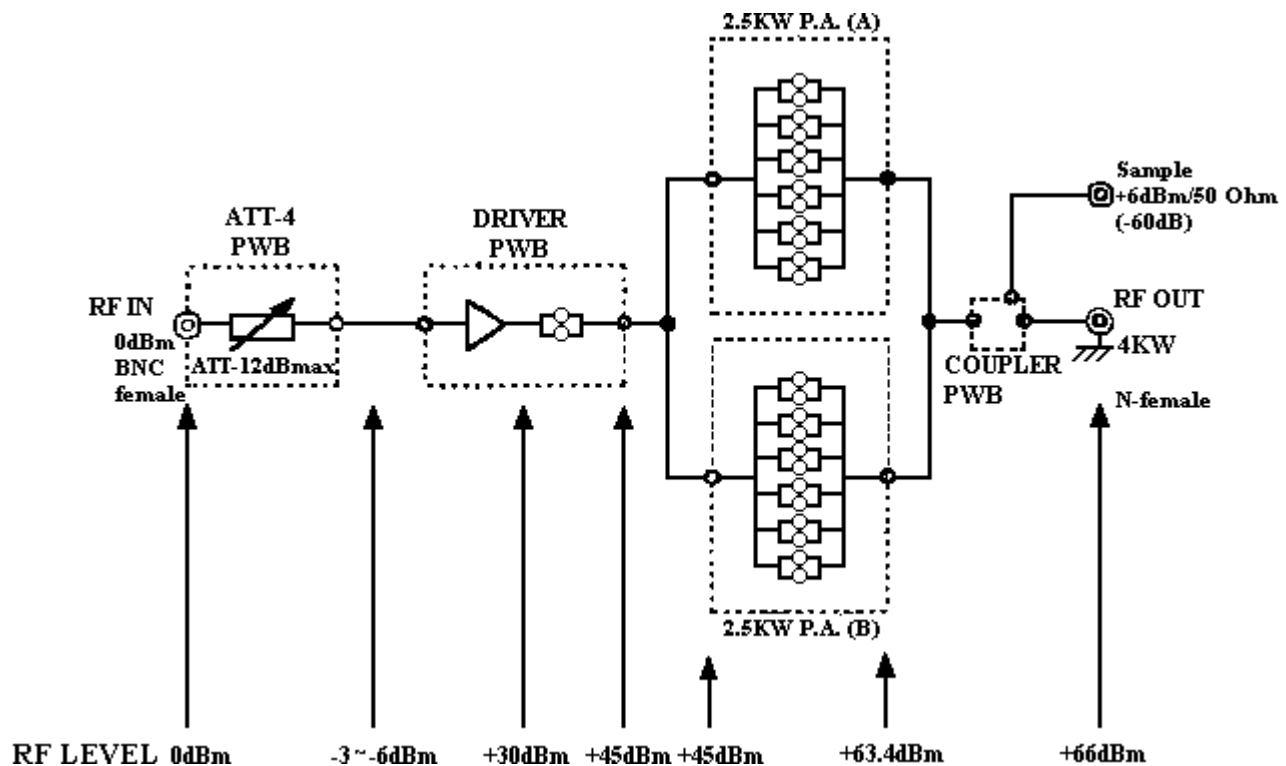
2-1 Fundamental Configuration Of RF AMP

Inner configuration of RF AMP is divided into the following sections.

- A) RF Amplifier section
- B) Control section
- C) Power supply section

2-1-1 RF Amplifier Section

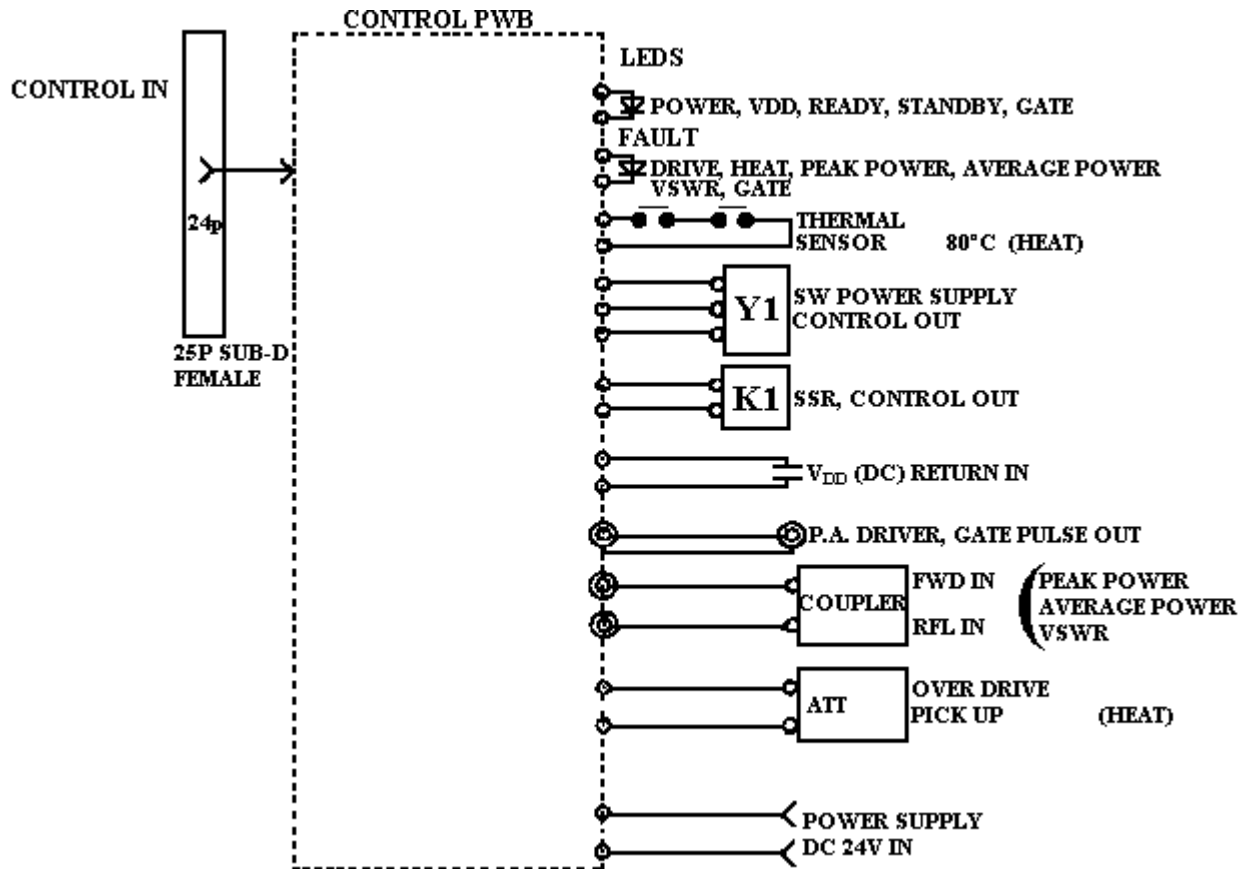
RF Amplifier section is composed of five PWBs. RF inputs are first received at ATT-4, then sent to Driver and amplified by approximately 50dB at this stage. Outputs from Driver are divided into two lines, and will be received at the two 2.5kw P.A. respectively. Each 2.5kw P.A. has amplification power of approximately 18dB. Power Mos FET, MRF150 (manufactured by Motorola) are adopted as the amplification devices. By operating 6 p.p. circuits (12p) of this device simultaneously, split and combine are made at the same phase, same level. 12p.p. combiner are performed on the two 2.5kw P.A., realizing 4kw.



RF AMPLIFIER LEVEL DIAGRAM
ILLUSTRATION 2-1

2-1-2 Control

This section functions as interface for external control signal, and controls various kinds of protective circuits, SW power supply, SSR. Messages are displayed on the front panel according to status code.



CONTROL DIAGRAM
ILLUSTRATION 2-2

2-1-3 Power Supply

AC primary power supply is;

AC 208V10% , Single phase, 50/60Hz, 20A

As for DC power supply, the following is used.

Y1 SW power supply for V_{DD}	1 unit
Electrical capacity	1500W
Output voltage	55V

Y2 SW power supply for control circuit	1 unit
Electrical capacity	15W
Output voltage	24V

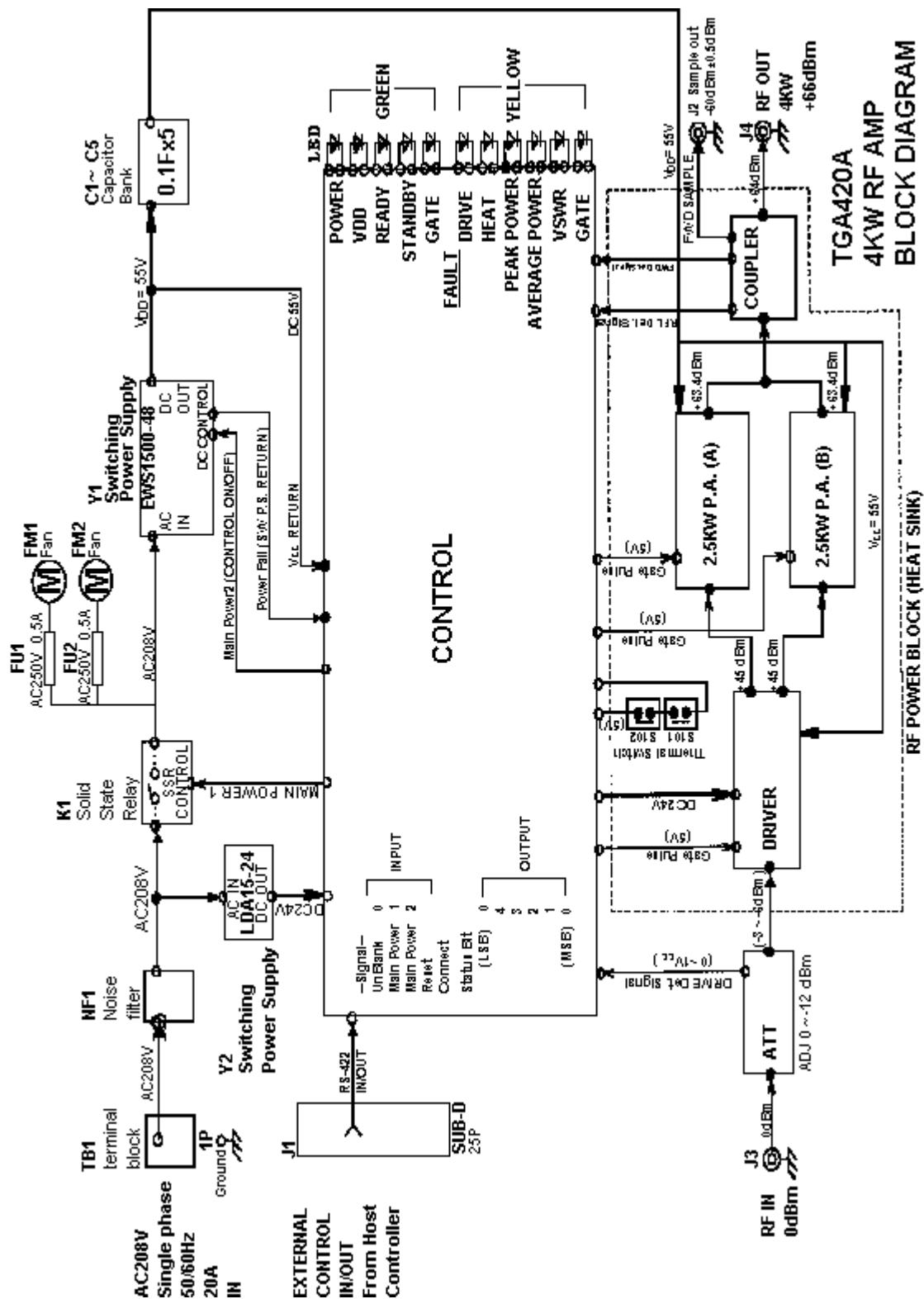
If the power is supplied from outside, Y2 starts, and then Control PWB starts. At this stage, only power LED on the front panel will light up, and Fan remain at rest.

After external signal main power 1 turn ON, SSR also turn ON, and Fan starts to turn.

After external signal main power 2 turn ON, DC output of Y1 also turn ON, and system becomes "ready" status.

Electrolytic capacitors (0.5F) are fitted to the output side Y1 (C1 to C5). After the DC power is turned OFF, some electrical charges linger at C1 and C2. It would take about 20 minutes for these remaining electrical charges to be discharged completely.

Initial set value of output voltage of Y1 by manufacturer is 48V. This has been readjusted at our factory.



RF AMPLIFIER BLOCK DIAGRAM
ILLUSTRATION 2-3

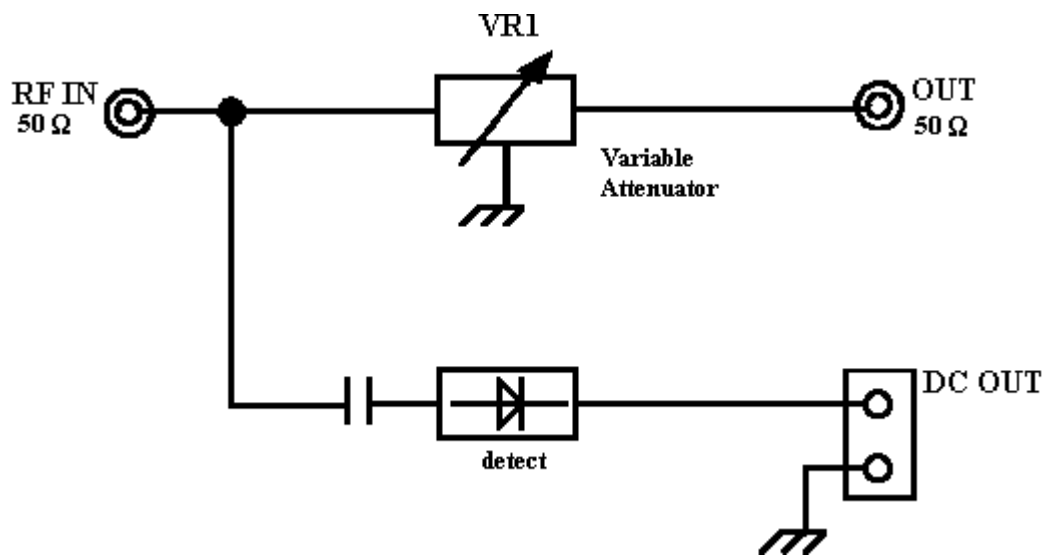
3- OPERATION OF EACH PWB

3-1 ATT-4 (PWB No. Q4B13857)

Total GAIN of RF AMP are variable by adjusting ATT (VR 1) (012dB seamlessly variable). Adjusting hole is located on the front panel. Adjustment of ATT is able to be done through this hole by using screw drivers. (Setting of Total gain has already been done at our factory before shipment. Normally, it is not recommended to touch this adjuster.)

Impedance of in-output is 50.

Part of RF IN level is sampled out for detection. This signal is used as detection-signal for over drive.



RF IN SIGNAL DIAGRAM
ILLUSTRATION 3-1

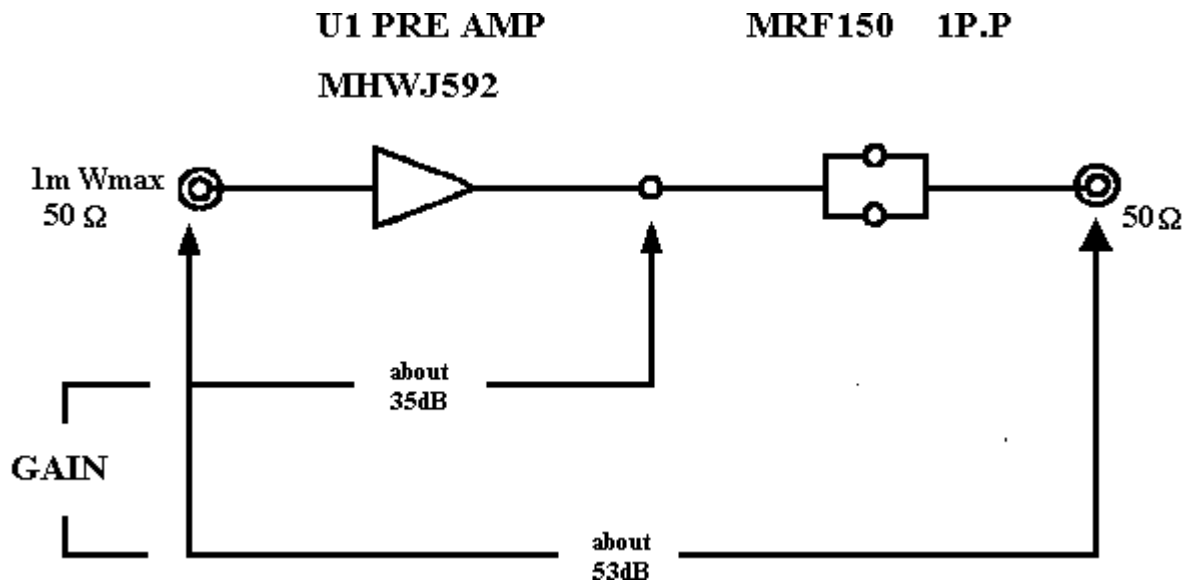
3-2 Driver (PWB No. Q2B13738)

Driver consists of two amplifying stages. For PRE-AMP stage, RF Module MHWJ592 (Made by Motorola) is adopted. The characteristics of PRE-AMP is;

Frequency band : 1-250MHz Power GAIN=35dB P out=0.8W Max.

For driver stage, Power Mos FET MRF150 is used. (A B grade 1P.P.)

Maximum input level is 1W, and Total GAIN is approximately 53dB.



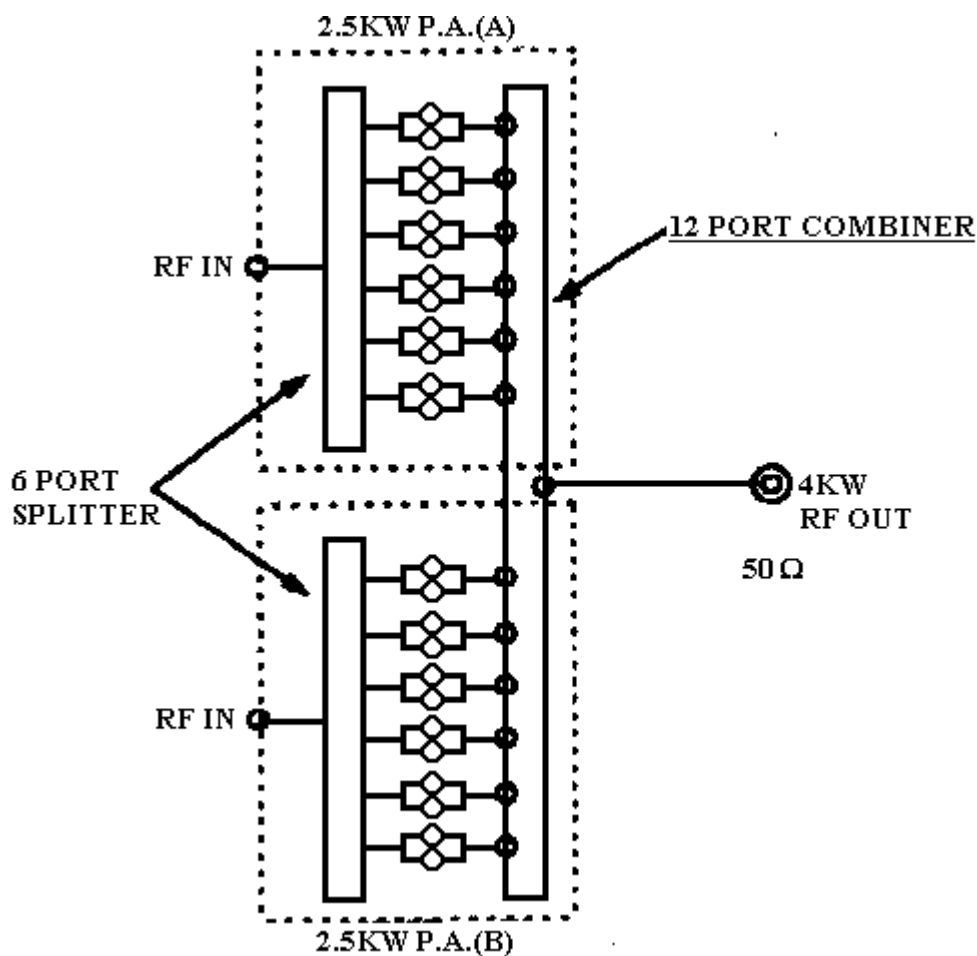
DRIVER DIAGRAM
ILLUSTRATION 3-2

3-3 2.5kw P.A. (PWB NO. Q1B13731)

For amplifying device, power MOS FET MRF150 (manufactured by Motorola) is adopted. This device is used as Class-A.B. 1 P.P. circuit to function as basic amplifying circuit. Each 2.5kw P.A. contains six of this basic amplifying circuit. AMP GAIN of 1PP circuit is 18dB.

RF input signals are split into signals of same phase and level by six port splitter. These signals will be input into each 1.P.P.circuit.

One unit of 4kw RF Amplifier have a pair of 2.5kw P.A. Outputs from each 1 P.P. circuit (62=12) are combined at the same phase and level by 12 port combiner to create 4kw RF output.



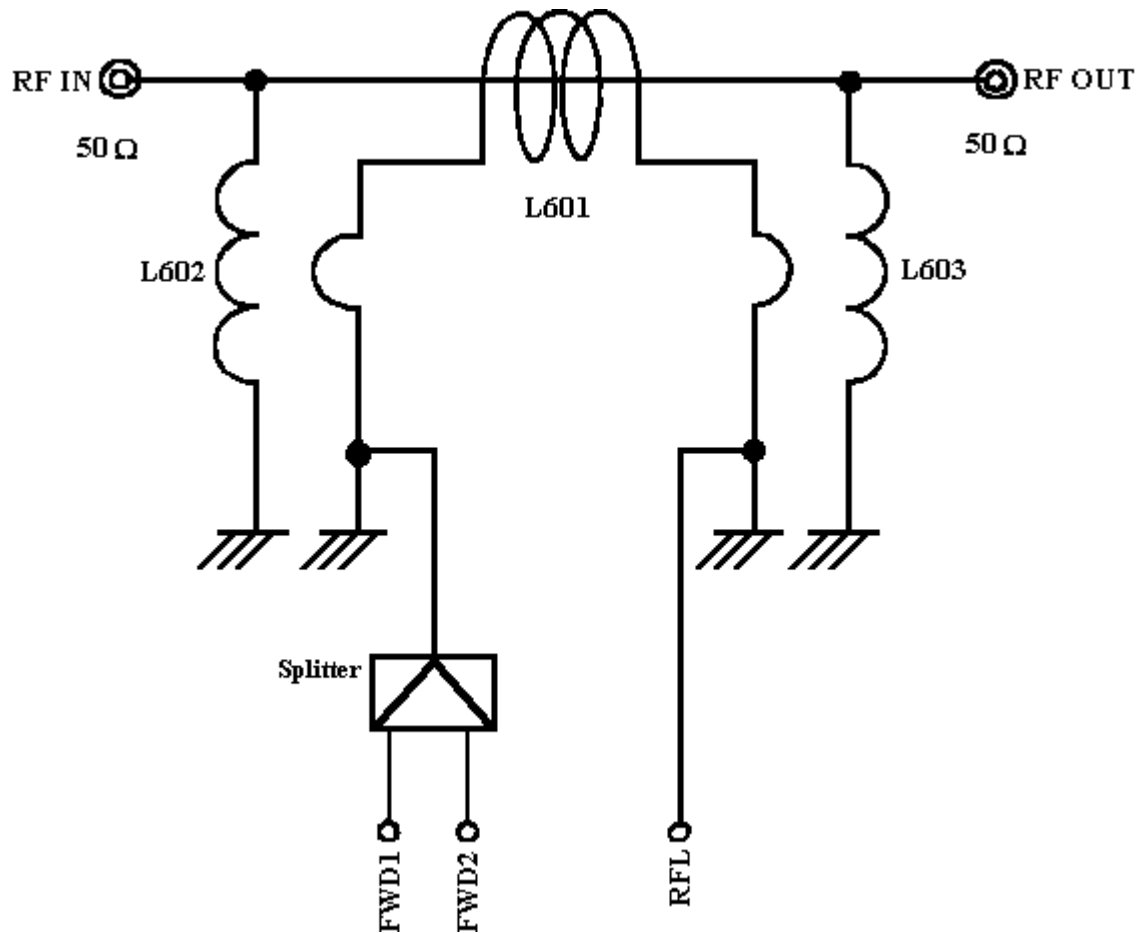
2.5KW P.A. CIRCUITS DIAGRAM
ILLUSTRATION 3-3

3-4 Coupler (PWB NO. Q3B13739)

This circuit detects forward-wave-power (FWD) and reflected-wave-power (RFL) of RF AMP output. High-frequency current is detected at coil L601, while high-frequency voltage is detected at coil L602, L603. Then, detected FWD and RFL are connected to Control PWB to be used for detecting protective circuit of peak power, average power, VSWR.

FWD is split into two lines, and connected to the front panel of RF SAMPLE (J2). Output level of RF SAMPLE OUT is 60dB0.5dB.

Output impedance is 50. Termination of 50 is essential.
Isolation of FWD and RFL is more than 20dB.



COUPLER DIAGRAM
ILLUSTRATION 3-4

3-5 Control (PWB NO. Q2M13741)

3-5-1 General

Control PWB has the following function.

- (a) Interface with the external control signal
- (b) Various kinds of protective circuits
- (c) LED display on front panel (status, protective circuits)
- (d) Control of RF AMP (AC 208V, SW power supply, GATE)

RS422 is used for external control signal. P.L.D. is adopted for logic circuit of Control PWB. Initial version of P.L.D. is Ver. 1.00.

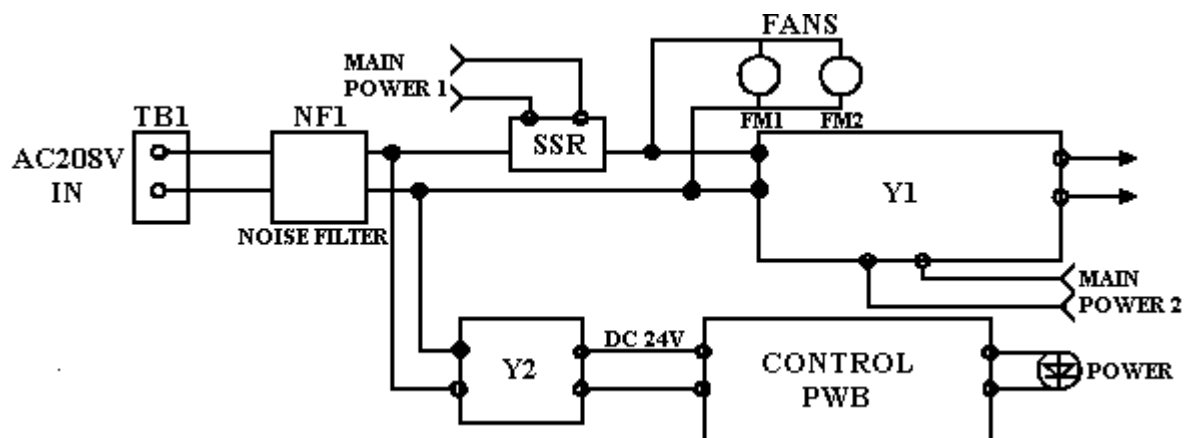
Ver. 1.00

Changes of version conform to following notational rule.

Minor changes in part of P.L.D circuit the number in digit goes up as these change occurs. (01,02)

Major changes (including changes in the number in digit goes up as these pin assignment, etc.) change occurs. (1,2)

Supplied voltage to Control PWB is DC24V. This is modified into 15V and 5V in PWB. When AC208V is supplied into RF AMP, Control PWB becomes operational condition. At this point, power LED (Green) on the front panel lights up.



VOLTAGE TO CONTROL PWB DIAGRAM
ILLUSTRATION 3-5

3-5-2 Interface

1. Signals for interface are shown below.

- A) Unblank 0 This signal is unblank signal in synchronization with RF signal. Gate control of each FET of Driver and 2.5kw P.A. are performed by this signal. In RF output mode, FET GATE is ON, while in RF Non-output mode, FET GATE is OFF. Under the condition that each protective circuit is in operation, FET remains OFF to reduce AMP GAIN even if there is RF input. In GATE OFF mode, this signal serve another function to reduce thermal noise level. In the case that RF output is single output (0.2T, 1.0T), Only this signal will be used.
- B) Unblank 90 This signal will be used when RF signal is dual signal like 0.5T type. Under dual output condition, two of RF AMP are able to be GATE controlled separately.
- C) Main power 1 When this signal is ON, SSR interposed in AC208V line comes ON. Then FAN starts, and power is supplied to Y1.
- D) Main power 2 This signal is for setting DC OUT of Y1 (EWS1500-48) ON/OFF. Y1's DC OUT ON/OFF circuit is controlled via this signal.
- E) Reset After inner protective circuits of RF AMP operates, that condition of protective operating will become "HOLD". To release this, This reset signal is used. (However, if this reset is used without removing the operational causes of the protective circuit, this reset function fails to work and starts protective operation again)
- F) Connect This signal is for confirming the connection of control connector. The number 6 and 19 pin in the D-sub 24p connectors are made short.
- G) The above signals are external input signal into RF AMP. With the exception of (f), all others are received at photocoupler.
- H) Status This signal outputs the status of RF AMP in six bit status code. (Please refer to Operation manual for details, including code table) These signals are output in AM26LS31CN(TI).

3-5-3 Protective circuits

Protective circuits have the five kinds of protective functions shown below.

These protective circuits have the detector circuits for operation. The basic operations of AMP protective circuits are as follows.

The detected signals via detector circuits (DC) are input into comparator. Threshold level of comparator is able to be set by volume. If DC level exceeds this set level, flip-flop circuit starts. In this way, irrespective of external control signal, FET GATE (Unblank), SSR (Main power 1), Y1 (Main power 2) are forcefully set OFF.

A) Drive

Drive operates by RF IN detector signal detected via ATT PWB. Comparator threshold setting is done by VR1. The setting point is set to operate within the range of 3dBm to 0.5dB of RF IN level.

B) Heat

Two thermal switches are fitted in series on the heat radiating board on P.A. The set temperature of thermal sensor is 80°C. Contact point of thermal sensor is normal-close. If temperature exceeds 80, Contact becomes open, and protective circuits start.

C) Peak power, Average power

These operate by forward-wave (FWD) detector signal detected via coupler.

Peak power This shuts off output at the absolute value level of RF output. Operating output level is within the range of 66.5 to 67.5 dBm. Setting can be made through VR3.

Average power RF AMP outputs in the pulse train. If average power of pulse output reach certain level, this protective circuit shuts off output.

This protective circuit operates if the average value of RF output pulse is within the range of 450W to 475W. Setting can be made through VR11.

D) VSWR

This operates by reflected-wave (RFL) detector signal detected via coupler. This protective circuit starts to operate, and then shuts off output, if load-impedance under 0dBm of RF input level exceeds 250 (VSWR4). The threshold point of this circuit is determined at the rated input level. Therefore, under low input level, there will be cases that this protective circuit fails to function even if load-impedance exceeds 250. This is not due to the malfunction. Setting can be made through VR5.

E) GATE

This protective circuit starts when the pulse width (ON time) of unblank input from outside falls between the range of 30 to 35m sec. Setting can be made through VR7.

The above are included in the protective circuit. This PWB also contains one unit of identical circuits for each function for 0.5T (dual) use.

3-5-4 Display on the front panel

- A) Power—This shows that AC 200V is supplied. Circuit-wise, this lights up when 24V power of Y2 comes ON.
- B) Vdd—This lights up when DC output of Y1 comes ON. There may be cases that electrical charges remain charged in the electrolytic capacitors (C1 to C5) on the inner circuits even if this LED has already dimmed out.
- C) Ready External output signal—in sync with Ready
- D) Standby External output signal—in sync with Standby
- E) Gate—This illuminates up when FET GATE is ON, dims out when OFF, and blinks, under normal output condition. This is not in sync with external control signal nor unblank signal.
- F) The all above displays are done by GREEN LED.
- G) Fault—This illuminates up under protective circuit operation described in section (4-5-3). The color of LED is yellow.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
0	June 30, 1998	TELI (KK)	Initial Release.
1	Oct 13, 1999	M. Keber	Added correct proprietary heading to document.